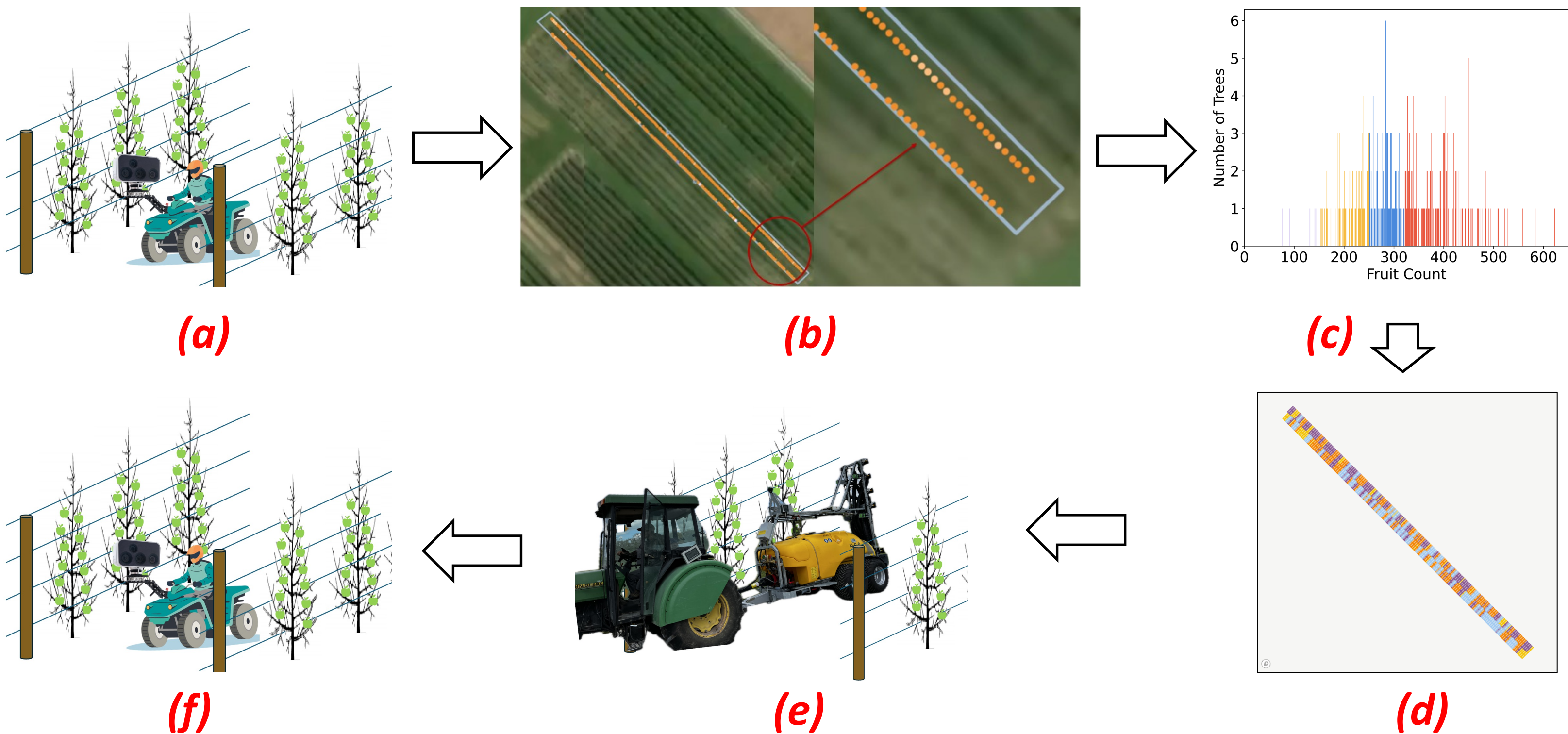


Integrating Computer Vision and Precision Sprayers for Targeted Green Fruit Chemical Thinning



Chenchen Kang^{1,2}, Hanjie Dou^{1,2}, Long He^{1,2}, Shanthanu Krishna Kumar^{1,3}
¹ Fruit Research and Extension Center, The Pennsylvania State University, Biglerville, PA
² Department of Agricultural and Biological Engineering, The Pennsylvania State University, University Park, PA
³ Department of Plant Science, The Pennsylvania State University, University Park, PA

Overview of the integrated orchard thinning workflow



- a:** Orchard is scanned to detect and count fruitlets.
- b:** A georeferenced fruitlet map is generated.
- c:** Tree-level spray needs are calculated.
- d:** A prescription spray map is created.
- e:** Sprayer adjusts rates based on fruitlet counts.
- f:** Orchard is scanned again to evaluate the remaining fruitlet load.

Computer vision



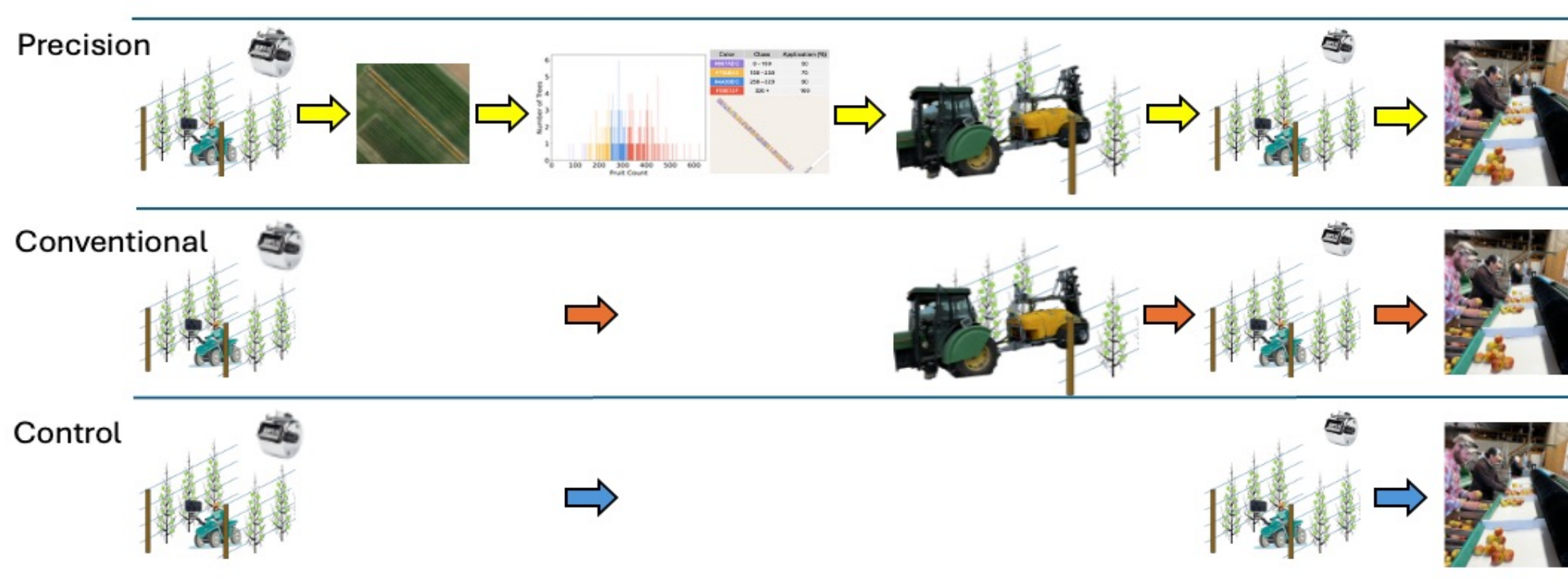
Precision sprayer



2024 “Fuji” trial at FREC



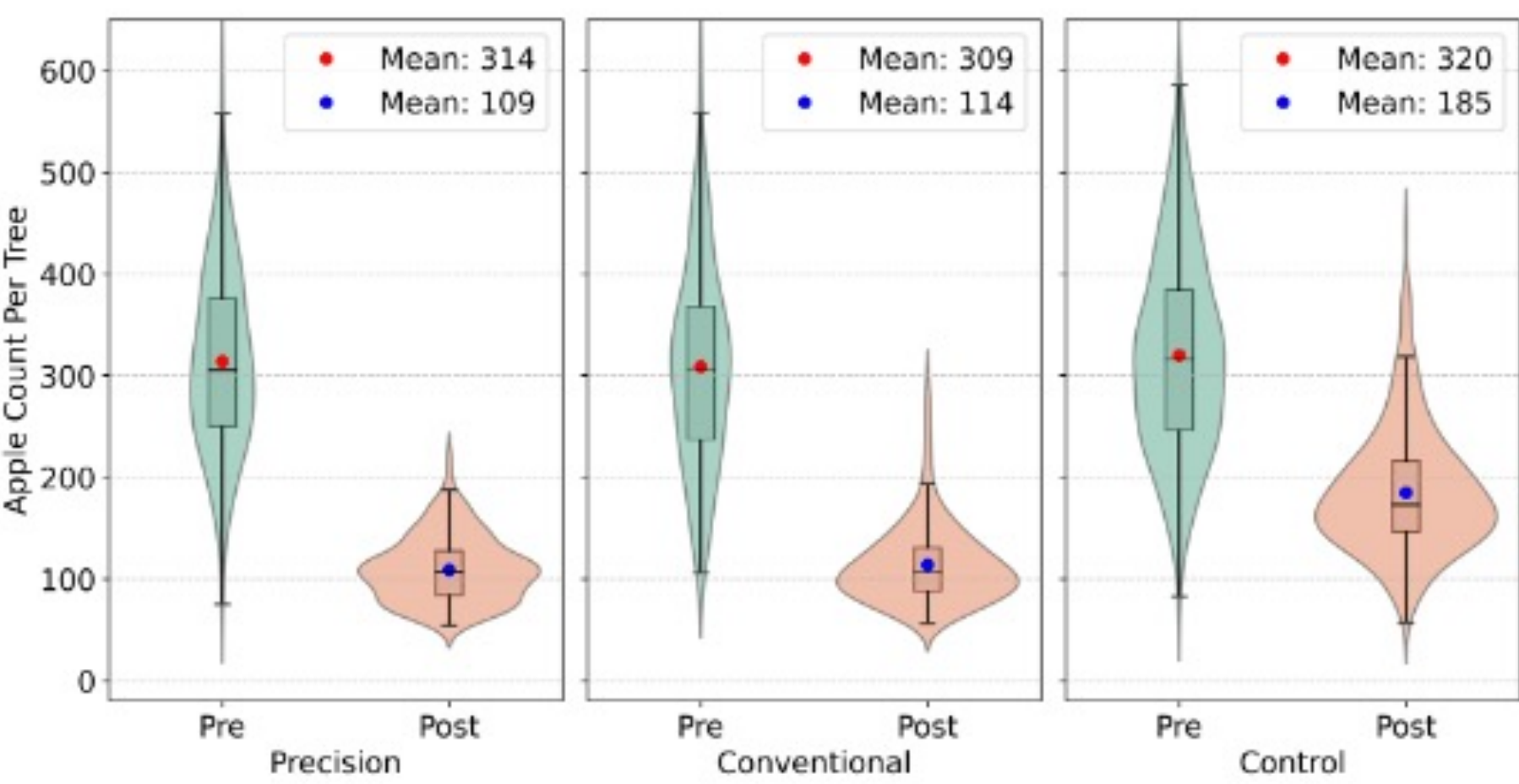
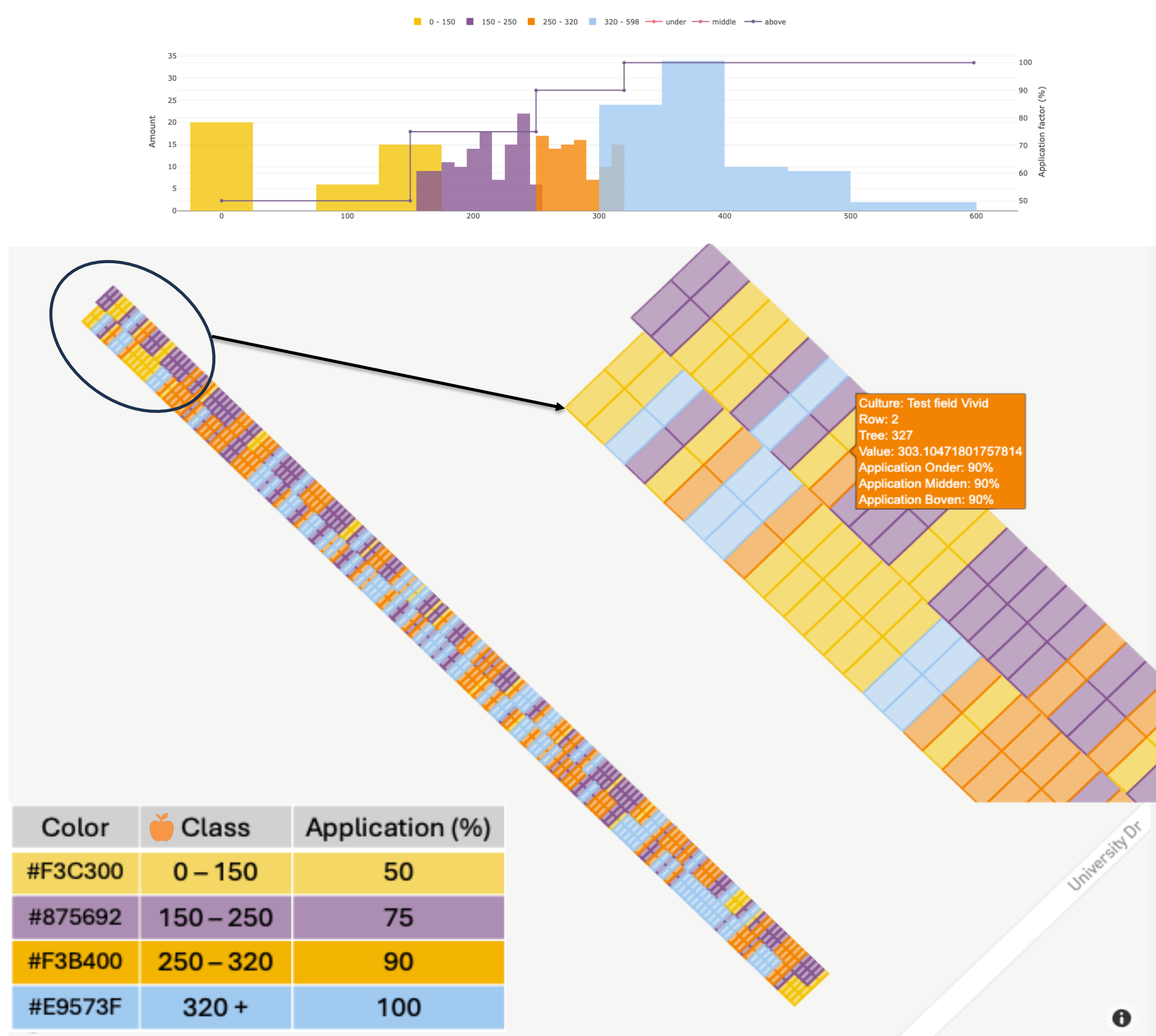
- 1:** Two rows received the integrated precision thinning (**precision**) treatment.
- 2:** Two rows were thinned using a uniform application rate (**conventional**).
- 3:** The remaining two rows were left untreated (**control**).



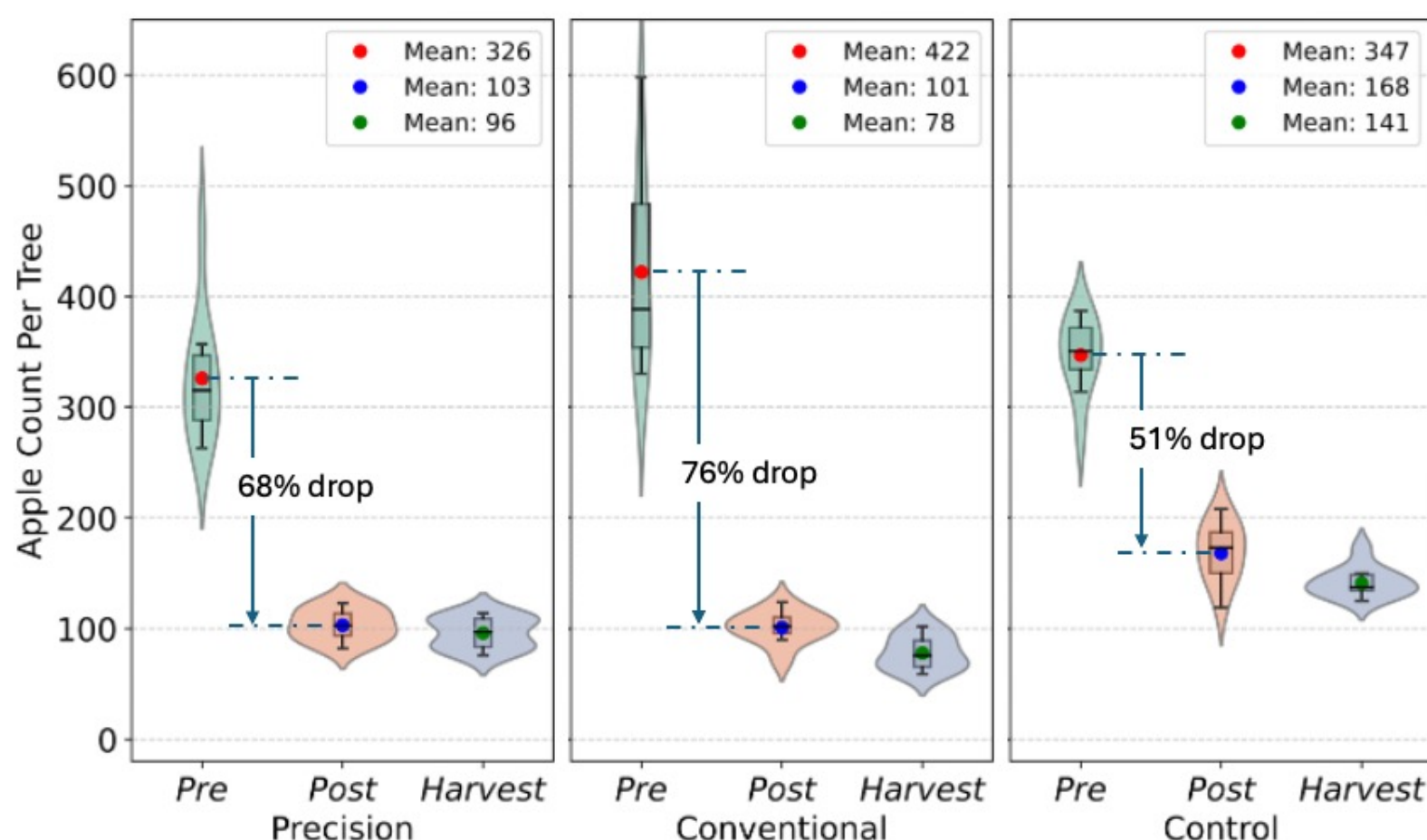
82.15% of the conventional chemical application
91 gal/acre

Each group (precision, conventional, control) was first scanned with Vivid XV3 for fruitlet counts, then received its respective treatment (or none), followed by a second scan and final harvest evaluations for yield and quality.

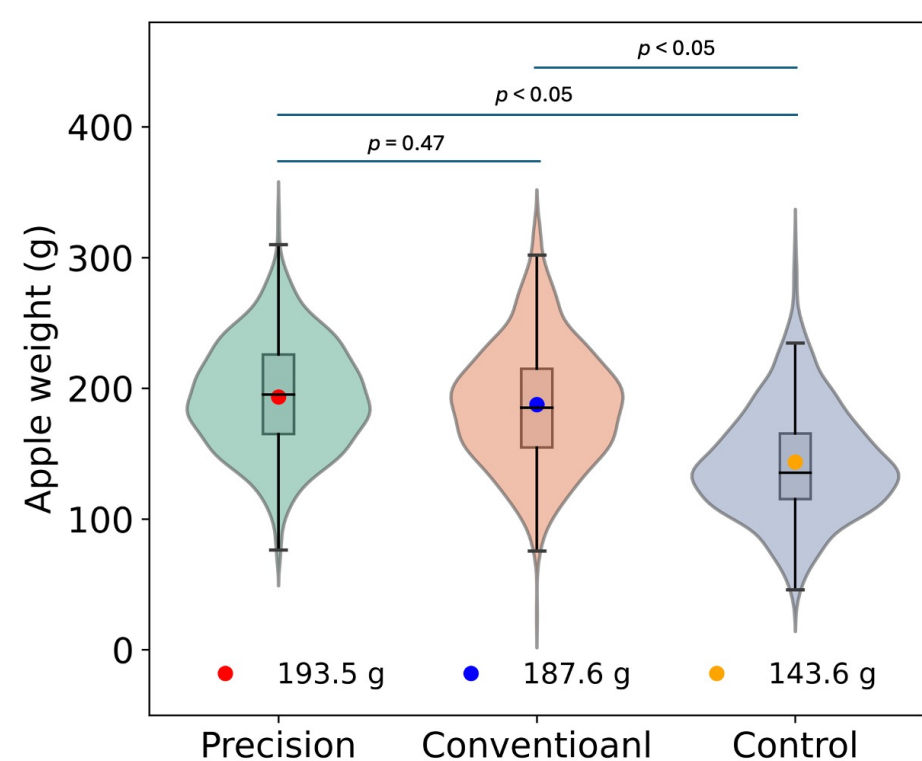
Detailed spray map for two rows of precision treatment



Fruitlet counts before and after thinning across three groups (all trees) with the **computer vision platform**.



Manual fruitlet counts from 10 trees in each group before thinning, 17 days after spray and at harvest.



Average mass of harvested apples

2025 ongoing trial at ACN

2025 Larger-scale trials in a Honeycrisp orchard

P: Precision, variable-rate
C: Conventional, fixed-rate

